

An Historical Preface to Engineering Ethics

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Abstract: *This article attempts to distinguish between science and technology, on the one hand, and engineering, on the other, offering a brief introduction to engineering values and engineering ethics. The method is (roughly) a philosophical examination of history. Engineering turns out to be a relatively recent enterprise, barely three hundred years old, to have distinctive commitments both technical and moral, and to have changed a good deal both technically and morally during that period. What motivates the paper is the belief that a too-easy equation of engineering with technology tends to obscure the special contribution of engineers to technology and to their own professional standards and so, to obscure as well both the origin and content of engineering ethics.*

Keywords: engineering ethics, science, technology, codes, values

The words “science”, “technology”, and “values”, like “engineering” and “ethics”, have been used in enough different ways to be dangerous. In this article, I will distinguish the most important uses of these five words, set them in context, and explain how they are (or are not) related. This conceptual house-cleaning is, I believe, a necessary prolegomena to productive work in engineering ethics (whether by philosophers, engineers, sociologists, or others). And, necessary or not, it has a rewarding byproduct, a foray into the history of engineering.

Foraging in another’s field is always risky. One can easily get lost, fall into traps the owners long ago learned to skirt, or find oneself suddenly outnumbered and outgunned. I am taking the risk for four reasons. First, I believe that reading history can lead to philosophical insights. The past gives the present context (as I shall try to illustrate here). Second, I believe that some historians, those I have been reading, sometimes miss the obvious—or, at least, get the emphasis wrong—and therefore tend to mislead those trying to understand engineering. I believe I can do better. Third, though I will be trespassing, I have precedent on my side. Philosophers have long made themselves useful by pointing out the obvious in fields not their own—which is all I intend to do. Fourth, and most important, I believe that my trespass will pay-off. Understanding engineering better, we shall be better prepared to understand what distinguishes its ethical problems both from those of other professions generally and from the sciences in particular.

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I. *TECHNE* AND *SOPHIA*: TWINS ANCIENT BUT UNEQUAL

I begin with etymology. "Technology" is a compound of two words from ancient Greek, "*techne*" and "*logos*". "*Techne*" means manual art. So, for example, a "*tekton*" was a carpenter or builder; an "*architect*", a master builder. The suffix form of "*logos*", "-ology", means a putting into words, an explanation or study. So, when our word "technology" still meant what Greek tells us it means, technology was the explanation or study of manual art, just as biology is the explanation or study of *bios*, life. It was a field in which gentlemen entered the workshop to record the artisan's secrets for later publication.¹

That, of course, is not what "technology" means now. Despite its Greek root, "technology" is really a new word, re-coined in the middle of the last century for a new idea.² What idea?

Ancient Greece was a slave-owning society and, like other slave-owners, Greeks tended to associate manual labor with slaves. Since no free man would want to be mistaken for a slave, the ancient Greeks generally avoided doing what slaves do. For example, because slaves tended to rush about on their master's business, free men were supposed to walk slowly.³ Greeks had such a low opinion of manual labor that they even rated sculpture less noble than painting because the sculptor, unlike the painter, had to sweat over his work like a slave.⁴

There were a few exceptions to this low opinion of manual labor. One was athletics. Athletics, however sweaty, was not something slaves did. War was another exception. Hacking one another with swords, though hard and dirty work, was a job for free men.

The Greeks contrasted *techne* with *sophia*. Though often translated as "intellectual knowledge" or even "science", "*sophia*" is probably better translated as "wisdom". From "*sophia*" comes our word "philosophy" (the love, that is, the pursuit of wisdom). For the Greeks, philosophy included mathematics, physics, economics, and similar sciences. Because philosophy was primarily a matter of thought, not manual art, philosophy was appropriate to free men.

The Greeks of Greece's Golden Age loved *sophia*; and she rewarded them accordingly. The Greeks of that period can claim credit for beginning the tradition of philosophy now dominant over most of the world, the one to which I belong. They can also claim credit for beginning a number of the sciences, including geometry, biology, and political science.

Their achievements in poetry, architecture, and history are no less impressive. Not so their contributions to *techne*. Of course, there were some contributions, for example, improved design of war galleys. But you must hunt for them. Europe's Dark Ages seem to have given us many more useful devices than did Greece's Golden Age.⁵

By now, perhaps, you can see two reasons to distrust that ugly word, "technology":

First, there is the implicit opposition between *sophia* and *techne*. Today we think of science and technology as related, not opposed. So, for example, one reason politicians give for funding scientific research is that it will pay off in new technologies.⁶

Second, there is the word's meaning in Greek. For us, technology is not—as its Greek parts suggest—a study of manual art but, primarily, our way of referring to all those inventions that make manual labor easier, more productive, or unnecessary. In this sense, technology began with the first tool someone made; the new technologies we hear about are new technologies in this sense, new tools someone has made. The academic discipline called "history of technology" studies the history of tools (together with ambient social arrangements).

Of course, there is yet another sense of "technology", one derived from this second

but referring to a study—as in, for example, the title “institute of technology” (or “technological university”). An institute of technology is not, as the Greek suggests, a place to study manual arts (carpentry, machining, and so on), a mere technical school. An institute of technology is, instead, a place to study practical inventions, how to make them and how to organize them (and those who use them) to make other useful things. The Greeks, who had a word for almost everything, seem to have had no word for that, technology as “the science of invention”.

So what? What does this history have to do with us? Consider, for example, how we dress for work: Some of us dress in “white collars”, that is, fine shirts, ties, good slacks, dresses, sport coats, or the like. Others wear “blue collars”, that is, coarse shirts, denim pants, coveralls. Generally, those in white collars have higher status than those in blue. Salary is secondary (as is social usefulness). A carpenter has less status than an accountant earning half as much. Why? Though carpentry requires a trained mind, it requires as well, like other blue-collar work, much sweaty labor surrounded by dust and debris. Because such labor would quickly ruin good clothing, the white collar guarantees some distance between its wearer and such “slavish labor”. And, because it does that, the white collar confers status.

No matter the origin of our parents, we are, in this respect at least, all more or less descendants of the ancient Greeks. Even if we ourselves like manual labor, we do not respect it as much as mental labor. I doubt that this is good, especially for engineers. But it does seem to be a stubborn fact about us. We are prejudiced against blue collars, not only those who work in them but even those who associate with those who work in them.⁷

That prejudice shows up even in a phrase seemingly having nothing to do with it—“science and technology”. Why does “technology” always come second? The explanation cannot be historical. If “technology” refers to inventions making manual labor easier, technology is older than science by tens of thousands of years. And, even if the “technology” in “science and technology” refers instead to the systematic study of practical invention, technology would be no younger than science in the corresponding sense, the systematic study of nature. For, until quite recently, “science” included all systematic knowledge, whether of nature or invention, including even jurisprudence and theology.

Nor can the explanation of the inevitable priority of “science” be alphabetical order. Substitute “engineering” for “technology” and the order remains the same: “science and engineering”, not “engineering and science”. Nor can the explanation be practical importance. Technology bakes our bread; science only helps us to understand how. Nor can the explanation be mere accident. Accident would produce more variation. The order seems fixed: science and technology. Why?

The answer, I think, is that the order indicates relative status. Science has higher status than technology; hence, it gets first mention.

Well, shouldn’t science have higher status? After all, isn’t technology just applied science? Doesn’t science come first in the order of development? Doesn’t science lay down the law, like a master, while technology merely applies it, like a slave? Even engineers may be tempted to answer yes to these questions. But the answer is: No, technology is not merely applied science. Many inventions (and much systematic knowledge of practical matters) preceded the scientific discoveries that explained them.

II. SCIENCE, TECHNOLOGY AND ENGINEERING

One can understand the words “science” and “technology” so that they refer to comparable concepts. Science is explicit, systematic knowledge of how “nature” works;

technology, explicit, systematic knowledge of how to make useful things. Unfortunately, usage today is not so neat. Though the term "science" did once refer (primarily) to explicit, systematic knowledge of nature, its meaning has now shifted somewhat so that, today, it refers as much, or instead, to a social undertaking, "a voyage of discovery" (as scientists like to say), rather than merely to what they discover. In this sense, science consists of certain communities engaged in trying to understand how nature works.

Since "technology" refers only to our practical inventions, or to the study of how to make more, we lack a term comparable to this new sense of "science". What do we call communities that invent useful things or, at least, add to our knowledge of how to do it? "Technician" is wrong; a technician is an assistant, one who carries out routine work under direction of a scientist, engineer, architect, physician, or the like. "Technologist", though a natural choice, has not caught on; "applied scientist", though once popular with sociologists, natural scientists, and even some engineers, is now fading.

Why? I think the reason is that the great majority of people who would have to be called "technologist" or "applied scientist" already have a satisfactory name, "engineer".

I said "great majority". I meant it. The United States today has well over 2,000,000 engineers. That is more than all other "technologists" together. Most other "technologists" are either architects, chemists, physicists, biologists, physicians, computer scientists, or "mere inventors" (that is, inventors not belonging to any of the preceding occupations). The US has only about 135,000 architects, 388,000 natural scientists (including chemists, physicists, and biologists), 450,000 computer scientists, and 600,000 physicians.⁸ I have no figure for "mere inventors"; but, since most inventors seem to be engineers, there can't be many "mere inventors".⁹ The number of physicians contributing to technology also cannot be large. Most physicians are not in research or development, but simply provide health care. So, even assuming that most scientists are in technology, not pure research, engineers must outnumber all other technologists combined by at least two to one.

These numbers suggest an obvious solution to the problem of what to call all those who make technology: call them "engineers". But that would, I think, be a terrible mistake. Chemists, architects, physicians, biologists, and the like are not engineers. Understanding why they are not will help us to understand both the values inherent in most technology, the technology engineers develop, and the place of ethics in any technology. It will also help us understand the difficulties inherent in talk of "social engineering", "genetic engineering", or "software engineering" where there is "engineering" without engineers. Understanding why not all technologists are engineers will bring us to the heart of our subject. But it will require more history, though (mostly) history less ancient than before.

III. THE BEGINNINGS OF ENGINEERING

Professions, aping aristocracy, like to trace their origins back to ancient times. So, for example, the American Medical Association's *Principles of Medical Ethics* cites certain provisions of Hammurabi's Code (about 2000 BC) as the earliest known code of medical ethics.¹⁰ There is, of course, some truth in such going-back. The healers of ancient Babylonia resemble today's physicians in many ways. There are, however, many differences as well; and, for our purposes, the differences are more important. We will understand professions better if we start their history with the rise of modern markets about two centuries ago, the accompanying dissolution of the old distinction between "trades" and "liberal professions", and the slow emergence of something

new.¹¹ Even an old occupation can be a new profession. By 1850, especially in England, we begin to see the modern pattern. The professions are connected both with a formal curriculum, ending with an examination and certification of some sort, and explicit standards of practice, a code of ethics.¹² Admittedly, those creating this new pattern seem unaware of doing something new. But there can be little doubt that they misunderstood their own actions. Even some of the terms they used were new. For example, the term “medical ethics” seems to have been coined in 1803 by a physician, Thomas Percival, for a book he thought was on an old topic.¹³

What is true of most professions is true of engineering. False pedigrees abound. Some histories of engineering begin with the Stone Age, with the first tools. They confuse engineering with technology.¹⁴ Other histories begin more sensibly, recognizing that engineers generally do not do manual labor but prepare instructions for others to carry out. Since the first tool almost certainly pre-dates such a division of tasks, these histories begin much later, with the first projects large enough to have some people laying out a plan and others implementing it. They begin with the building of Stonehenge, the Pyramids, or some other wonder of ancient civilization.¹⁵

Though better than the first, this second way of beginning the history of engineering still has at least two embarrassing consequences. One embarrassment is that it makes architects (or “master builders”) the first engineers. This is embarrassing because engineers generally agree that architects today are definitely not engineers. Another embarrassment is that this way of telling the story makes a mystery of why our word for engineer comes from French, rather than Greek, like “architect”, and why the French have had the word for barely four hundred years. Generally, we have a word for anything important to us almost as soon as we have the thing. There are no significant *whatchamacallits*.

So, when I tell the story of engineering, I start four hundred years ago in France. Back then there were things called “engines”—but “engine” then simply meant a complex device for some useful purpose, a contraption showing intelligence in design, in short, a machine. The first people to be called “engineers” were soldiers associated with catapults, siege towers, artillery, and other “engines of war”. They were not yet engineers in the sense that concerns us. They were, rather, engineers in the sense that, even today, the driver of a locomotive is an engineer. They were engineers only in the sense that they operated (or otherwise worked with) engines.

Some soldiers are still engineers in something like this sense: they belong to an engineering corps. Though they do not know what engineers know, they are directly involved in works of engineering, though not precisely with engines of war, a term no longer in common use.

Four hundred years ago the armies of France were led by nobles, men on horseback, who learned war from their fathers or on the battlefield, or died in the attempt. The foot soldiers came with the nobles. Most were peasants or artisans who knew little of war until trained in camp. When the war ended, the army dissolved, each noble leading his own people home. In such an army, an “engineer” was usually a carpenter, stone mason, or other artisan bringing civilian skills to war.

When Louis XIV ended the regency in 1661, France still made war in this way. But, within two decades, France had a standing army of 300,000, the largest, best trained, and best equipped European fighting force since the Roman legions. This achievement was widely copied. To this day, most of our military words—from “army” itself to “reveille”, from “bayonet” to “maneuver”, from “private” to “general”—are French. “Engineer” is just one of these military terms.

Until 1676, French engineers were part of the infantry. But, in that year, the “engineers” were organized into special units, the *corps du génie*.¹⁶ This reorganization

had important consequences. A permanent corps can keep much better records than isolated individuals, can accumulate knowledge, skills, and routines more efficiently, and can pass these on more effectively. A corps can become a distinct institution with its own style and reputation. More than a group of proto-engineers, the *corps du génie* were, potentially, both a center of research in engineering and a training ground for engineers (in something like our sense), *officiers du génie*.

The *corps du génie* did not take long to realize this potential. Within two decades, it was known all over Europe for unusual achievements in military construction. When another country borrowed the French word “engineering” for use in its own army, it was for the activity the *corps du génie* engaged in.¹⁷ That was something for which other European languages lacked a word.

The *corps du génie* was not, as of 1700, a school of engineering in our sense; it was more like an organization of masters and apprentices. Indeed, strange as this may seem now, at that time neither France nor any other European state had a permanent military academy (in anything like our sense), much less a school of engineering. There was no settled curriculum for training officers generally or engineers in particular, or even a very clear idea that that was necessary. Only during the 1700s did the French slowly come to understand what they wanted from an engineering education and how to get it. But, by the end of the 1700s, they had a curriculum from which today’s engineering curriculum differs only in detail; they had also invented engineering.¹⁸

An army needs fortifications for protection, mines under enemy fortifications, roads to march on, and bridges to cross. Civilians need either the same things or other things requiring similar skills to build. So, in 1716, the French established another corps of engineers, the *corps des ponts et chaussées*, to build and maintain the nation’s bridges, roads, and canals (as important to the army as to commerce). This corps set up a school for training its officers, the first engineering school to survive long enough to matter. Like the military engineers, these civil engineers were admired all over Europe. Those who copied their method copied as well the name they gave it.¹⁹

What was their method? Engineers, military as well as civil, resembled architects in being able to make drawings for construction projects, develop detailed instructions from those drawings, and oversee the execution of those instructions. They nonetheless seem to have differed from architects in at least three ways.

First, engineers were much better trained in (what was then) the new mathematics and physics than the architects. They had the ability to consider systematically questions most architects could only deal with intuitively or ignore.²⁰

Second, because the strategies of engineering had their roots in the necessities of war, engineers paid more attention to reliability, speed, and other practicalities. So, for example, the systematic testing of materials and procedures in advance of construction was early recognized as a characteristic of engineers.²¹ At least in comparison, the architect seemed an artist, one for whom beauty claimed much of the attention an engineer would devote to making things work.

Third, to be an engineer was to have been trained as an army officer, to have been disciplined to bear significant responsibility within one of world’s largest organizations. Engineers were therefore likely to be better at directing large civilian projects than architects, most of whom would have had experience only of much smaller undertakings.

These three advantages tend to reinforce one another. For example, large projects not only require more planning in advance and more discipline in execution, they are also more likely to require better mathematical analysis and to justify extensive testing of materials and procedures. For this, and perhaps other reasons, civil engineers slowly took over much of the work that once would have been the domain of architects. They

were a new power in the world.

Early experiments in engineering education culminated in the *École Polytechnique*. Begun in 1794 as the *École des Travaux Publics* (the school of public works), it changed its name the following year, for the first time connecting engineering and “*techné*”. I don’t know why the French did this. The school never trained architects, much less artisans or mechanics. It was a school of engineering, deserving the “poly” only for offering preparation for many fields of engineering, military and naval engineering as well as civil.²²

The *École Polytechnique*’s curriculum had a common core of three years. The first year’s courses were geometry, trigonometry, physics, and the fundamentals of chemistry with practical applications in structural and mechanical engineering. There was a good deal of drawing, some laboratory and workshop, and recitations after each lecture. The second and third year continued the same subjects, with increasingly more application to the building of roads, canals, and fortifications and the making of munitions. For their last year, students were sent to one of the special schools: the school of artillery, the school of military engineering, the school of mines, the school of bridges and roads, the school of geographical engineers (cartographers), or the school of ships.²³

Engineers will immediately recognize this curriculum, especially, the four years, the progression from theory (or analysis) to application (or design), and the heavy emphasis on mathematics, physics, and chemistry.

The *École Polytechnique* was the model for engineering education for much of the nineteenth century.²⁴ The United States began using it very early. Our first engineering school was the military academy at West Point. By 1817, it had adopted much of the *École*’s curriculum, its methods of instruction, and even some textbooks.²⁵

IV. VALUES IN ENGINEERING

What values does engineering incorporate? A decade ago, Eugene Ferguson, an engineer turned historian, drew up a list of what he called “imperatives of engineering”.²⁶ The list is neither complete nor fundamental—nor, indeed, even entirely fair. It will nevertheless help us understand engineering.

Engineers, Ferguson claimed, (1) strive for efficiency, (2) design labor-saving systems, (3) design control into the system, (4) favor the very large, the very powerful, or (in electronics) the very small, and (5) tend to treat engineering as an end in itself rather than as a means to satisfying human need. These “imperatives” are, according to Ferguson, instincts engineers bring to their work. While an engineer can resist them, just as I can resist drinking water even if I am thirsty, they are, in effect, the engineer’s default setting, what engineers will do unless they consciously try to do something else.

Ferguson intended this list to be a criticism of the way engineers work. It is, I think, both less and more than that. The list is less than a criticism because the first four imperatives seem, on reflection, at least as much virtues as vices. The list is also more than a criticism because it highlights certain enduring features of engineering, permitting us to connect engineering’s history with today’s practice. Let’s take a closer look at Ferguson’s list.

“Efficiency” is the first imperative Ferguson identifies. Ferguson points out, rightly, that “efficiency” is a slippery term, meaning “most powerful” here, “lowest cost” there, and something else elsewhere. What he overlooks is the concept’s utility.

Engineers generally define “efficiency” so that they can measure it (or its components), assign numbers, and thereafter seek to control it. That is not surprising. Like other professions, engineering tends to analyze a situation so that its distinctive

skills can be applied. One distinctive skill of engineers is giving mathematical structure to practical problems. The concept of efficiency allows them to exercise that skill.

Engineers have, no doubt, sometimes paid too much attention to efficiency, especially forms of efficiency that turned out not to matter. Indeed, the history of engineering is in part the history of measurable properties used for a time as proxy for something that could not be measured and then discarded when the proxy proved not to have enough of a relation to what the engineers actually cared about.²⁷ Because engineering is a practical undertaking, it must learn from practice. It cannot learn from practice without making mistakes. Some of engineering's mistakes have concerned efficiency.

Engineers can be unduly slow about giving up one of these proxy measures. But, even this slowness is understandable. Engineers are used to working in large organizations, organizations where change is difficult and the consequences are often hard to predict. They therefore have a tendency to follow practices they would no longer adopt. (Consider, for example, how engineers still specify 1/4 inch bolts or screws rather than their metric equivalents.) The world is a tough laboratory. Many things better in theory are worse in practice. How daring do we want engineers to be with our lives?

The second imperative on Ferguson's list is a preference for labor-saving devices. Engineers will, Ferguson thinks, design to save labor even when labor is cheap and the end result will be higher production costs and more unemployment.

The engineer's preference for labor-saving is understandable as a product of engineering's military origin. Since engineering began, the primary labor pool of most armies has been their own soldiers. Since no general wants his soldiers doing construction when they could be fighting, military engineers have always had an incentive to look for means of saving labor even though the labor saved was, in one sense, cheap (indeed, free).

As military engineering became civil engineering, this tendency might have put engineers at a disadvantage. Their designs might have proved too costly. Those who hired engineers would, however, soon have learned this. They would then have compensated, either by calling in an engineer less often or by making sure that the engineer called in defined the desired outcome taking cost into account.

If, as Ferguson's criticism suggests, such compensation seldom occurs, the most likely reason is that the engineer's preference for labor-saving devices generally serves those who employ engineers. The reason that preference might serve their employers is not hard to see. Labor has a tendency to become scarce, and so costly, where it is not routinely saved.

Of course, that is only a tendency. Many of those thrown out of work by a particular innovation may end their lives on the dole. Many engineers would, no doubt, like to take such bad effects into account; and perhaps many of their employers would let them. But, if engineers are to take such considerations into account, they will need both the relevant information and a routine for using it.

Gathering such information belongs to the social sciences, not to engineering as it is or as it is likely to become. Any curriculum that could give engineers the skills to develop significant social statistics would probably be too long to attract many students. Engineers should not be blamed for failing to take into account social consequences about which they can only guess.

Where, however, such information exists, developing ways to incorporate it into engineering work is certainly something engineers can, and should, do. Indeed, they have long done this with the employer's share of the cost of production. And, over the last two decades, thanks to the Environmental Protection Agency (EPA), engineers

have become adept at incorporating environmental costs into their designs (for example, by designing for disposal as well as for manufacture and use). They could do the same for social impact if they had numerical standards for assessing impact and sources of information from which the relevant numbers could be taken.

Engineers can help to develop such standards, just as they helped to write EPA standards. But, as with environmental standards, standards for permissible social impact are probably not what most people would want engineers alone to decide—or even engineers with the help of lawyers, accountants, corporate executives, and other specialists. Social impact raises political issues, that is, issues everyone wants a part in deciding. If engineers decline to develop such standards unilaterally, should we blame them?²⁸

Ferguson's third imperative is designing controls into the system. Engineers generally try to separate planning and execution. Intelligence is designed into the system, requiring as little intelligence as possible of the system's operators. The assembly line is the typical example of this imperative. Engineers have generally tried to design an assembly line so that the work is so simple that only a few minutes training is necessary to learn the job. The job is therefore likely to be repetitive and boring; those doing the job to be reduced to little more than organic robots.

Engineering's military past certainly explains the origin of this imperative. Soldiers sent over to help on an engineering project, whether digging trenches or putting a bridge over a river, will not have much time to learn the job. The military engineer must design the work so that anybody can do it.

But its military past alone does not explain why this imperative persists in civilian engineering (or, at least, why engineers who do such things should be so much in demand). The explanation of that, like the persistence of engineering's second imperative, must be that this tendency has proved useful in civilian engineering as well. One recent example will suggest why that might be.

McDonald's restaurants now have cash-register buttons with pictures of the various items on the menu. The cashier need not know the price of anything, or even be able to read, only be able to recognize the pictures and push buttons accordingly. In a business where employee turnover is high and education low, where prices change frequently and training is expensive, this dumbing-down of the cashier's job both saves money for McDonald's and opens employment to many who might not otherwise be qualified. Whoever thought of that device, engineer or not, was undoubtedly a hero to McDonald's.²⁹

The fourth imperative of engineering Ferguson lists is a tendency to disregard human scale, preferring the very large or the very small. The reason for this imperative is that engineering was, and remains, a creature of large organizations. Louis XIV's army, one of the largest organizations of its day, created engineering to do what civilian artisans could not do (or could not do well enough). Even today, most engineers work in large organizations. You do not need an engineer to construct a single family house. A carpenter or architect will do, as they always have. If, however, you want to construct a thirty-story building, you will need engineers.

The problem, I think, is not so much that engineers disregard human scale as that they are seldom needed for things on a human scale. Generally, asking engineers to work on a human scale is like asking lawyers to prepare a partnership agreement for two children opening a lemonade stand. They can do it, but either they will do what anyone else could do or they will do something out of all proportion to the job.

In this respect, the very small can be like the very large. For example, to make today's tiny electronic circuits requires productive forces and controls of which a single human being is incapable. Hence, there is work for engineers.

Ferguson's last imperative, putting technical brilliance ahead of human need, is unlike the others. It is a failing common to all professions.—We all know the joke about the surgeon who says, "The operation was a success, though the patient died."—But this last "imperative of engineering" is worse than a failing common to all professions; it is a failing inconsistent with one of engineering's fundamental values.

I have stressed the military origins of engineering.³⁰ I have not pointed out that most of the period we have been talking about, roughly the 1700s, is known as the *Age of Enlightenment*. This was the time when many Europeans first came to believe that enlightenment, that is, scientific learning, would bring peace, prosperity, and continuous improvement. For countless ages, the best hope of the wise was that the world would not get much worse. With the Age of Enlightenment, people began to act on the belief that the world could be made much better. Engineering has this belief built into it. So, for example, early graduates of the *École Polytechnique* were noted for "scientific and democratic idealism and a desire to work for human progress".³¹ The same attitude was evident in England at about the same time. When, in 1828, the British Institution of Civil Engineers, then nine years old, asked one of its members, Thomas Tredgold, to define the term "civil engineering", he gave an answer engineers still quote: "Civil Engineering is the art of directing the great sources of power in Nature for the use and convenience of man....The most important object of Civil Engineering is to improve the means of production and of traffic in states, both for external and internal trade."

For Tredgold, engineering was committed to making things "for the use and convenience of man". But, for Tredgold, this was not simply a matter of maintaining things as they are. Engineering was supposed to "improve means of production and traffic". Engineering was, by definition, an instrument of material progress.

But what about engineering today? Most engineers would, no doubt, want to tinker with Tredgold's definition, for example, by substituting "people" for "man". But few, if any, would want to change its core. Engineering remains an undertaking committed to human progress. So, for example, the most widely adopted of America's codes of engineering ethics, begins: "Engineers uphold and advance the integrity, honor, and dignity of the engineering profession by...using their knowledge and skill for the enhancement of human welfare."³²

V. WHY ENGINEERS ARE NOT SCIENTISTS

That is enough about engineering. We are ready to see how engineers differ from other technologists. I have already pointed out some of the ways engineers differ from architects.³³ I shall now explain how they differ from applied *scientists*.

I once did a workshop at the research lab of a large petroleum company. The audience was about half chemists and half chemical engineers. I first asked the chemists, "If you had a choice between inventing something useful and discovering new knowledge, which would you prefer?" The chemists thought this a hard question: "After all," they reasoned, "it's hard to imagine an interesting discovery in chemistry that would not have a practical pay-off." Eventually, I asked for a show of hands. About half the chemists voted for "something useful" and about half for "new knowledge". The engineers, on the other hand, *all* voted for usefulness. For them, new knowledge was just a means to improving human life.³⁴

Unlike the chemists, the engineers had no commitment to science as such. They used science, much as they used other sources of insight. They also contributed to science much as they contributed to lawyering and other social practices, for example, by helping to reduce the cost of chemicals used in the manufacture of computers. But they did this as non-scientists, as participants in a voyage of invention rather than

discovery.

This difference between engineers and chemists came as a surprise to these researchers. Many had worked side by side for decades. They thought that they shared the same values. But we do not wear our values on our clothing like an identity badge—except, of course, by declaring ourselves to be of this profession rather than of that. These researchers had not taken the difference in profession seriously. That is why they were surprised.

This difference between scientists and engineers is not a mere idiosyncrasy of one industrial laboratory. I have asked the same question at other industrial laboratories since, with much the same result. Nor is this difference between scientists and engineers restricted to industrial laboratories. I have asked my question at university workshops attended by both engineering and science faculty. The results were even sharper. The only thing rarer than a university scientist who voted for “something useful” was an engineer who voted for “new knowledge”.³⁵ In this respect at least, engineers are not scientists, not even applied scientists. The primary commitment of engineers is not to knowledge, theoretical or applied, as one would expect of scientists, but to human welfare.³⁶

VI. ETHICS AND ENGINEERING

Earlier, I described engineering as a “new power in the world”. Power, though neither good nor bad in itself, is always dangerous. Because of the scale on which engineers generally work, engineering is particularly dangerous. Engineers long ago realized this and set about to assure, as much as possible, that engineering would be used for good rather than evil. They organized as a profession, adopted standards of practice, including “codes of ethics”, and tried to put them into practice.

“Ethics” has at least three common uses. It can refer: 1) to ordinary morality; 2) to the systematic study of ordinary morality (for example, the attempt to understand morality as a rational undertaking); or 3) to those special morally-permissible standards of conduct every member of a group wants every other member to follow even if that would mean having to follow the standards too. It is in this third sense, I think, that members of a profession talk of their “profession’s ethics”.³⁷ A profession’s ethics are standards of conduct applying to them in virtue of their profession. Legal ethics applies to lawyers; medical ethics, to physicians; and so on.

In this sense, engineers did not have a code of ethics until they adopted one. The engineers’ code of ethics is what they make it—so long as the standards they lay down are consistent with ordinary morality.³⁸ That means that engineering ethics can change over time and even differ from country to country or field to field. Consider a particular provision of a particular code of engineering ethics, for example, the most cited of all: “Engineers shall hold paramount the safety, health and welfare of the public in the performance of their professional duties.”³⁹ This provision may now seem like the natural, perhaps even the necessary, expression of the engineering’s commitment to improve human life. But its history reveals it to be a controversial—and quite recent—achievement.

The first American codes of engineering ethics were adopted in the second decade of this century. None contained the paramountcy provision. In its place, we find provisions requiring, for example, that the engineer “be guided by the highest principles of honor”, that he “satisfy himself to the best of his ability that the enterprises with which he becomes identified are of legitimate character”, and “[if] any...professional obligations...interfere with his meeting the legitimate expectations of a client or employer, the engineer should inform [the employer].”⁴⁰

The absence of the paramountcy provision was no mere oversight. As early as

1922, Morris Cooke, a mechanical engineer, himself active in drafting one of the early codes, was pointing out in print “the failure [of engineering codes] to mention the public interest as a test—if not the supreme test of action”.⁴¹ Yet, more than three decades later (1947), the predecessor of the paramountcy provision merely required the engineer to “have due regard for the safety, life and health of the public and employees who may be affected by the work for which he is responsible.”⁴² By 1963, the corresponding provision required the engineer merely to “have proper regard for the safety, health, and welfare of the public in the performance of his professional duties.”⁴³ Only in the 1970’s was the paramountcy provision adopted.

If we compare these various provisions, we will not find a simple progress toward ever higher standards. The paramountcy provision, like its immediate predecessor, drops the reference to “employees” in the 1947 provision. It, in fact, seems to lower standards with regard to employees even as it seems to raise them with regard to the public. We do not know why these provisions changed as they did or when they did. Here is work for historians. What we can see is that each of these provisions seems to be a morally permissible standard of conduct expressing (something like) the same commitment to improving human life, though each in a different way and with a different emphasis.

There is, then, an important difference between (moral) values such as the engineers’ commitment to improving human life and ethics (special standards of conduct). While engineers generally seem to have valued human welfare since early in the history of engineering, failing to treat the public welfare as paramount in their professional work could not be unethical for engineers (in our third sense of “ethical”) until engineers adopted a standard of conduct requiring them to treat the public welfare as paramount. And, except perhaps for utilitarians, failing to treat the public welfare as paramount could also not be immoral. Morality imposes no such duty (though it does require us to avoid doing certain kinds of harm and perhaps to render certain kinds of aid). The history of engineering ethics reminds us that ethical standards, like other engineering standards, are not discoveries but useful conventions.

Values are reasons for acting, and so reasons for adopting standards of practice, but values as such, even moral or professional values, do not tell us how to act. A value as such can only tell us that this is something to consider. Standards of practice, including a code of ethics, do tell us how we should act. They deserve special attention in any discussion of engineering ethics.

NOTES AND REFERENCES

This article began as a GTE Lecture at the University of Wisconsin Center/Fond du Lac, 13 October 1992. I should like to thank those present, as well as my colleagues, Wilbur Applebaum (History) and Sid Guralnick (Civil Engineering), and my friend, Mike Rabins (Texas A & M), for help in sorting through the issues discussed here.

- 1 See, especially, Paolo Rossi, *Philosophy, Technology, and the Arts in the Early Modern Era*, trans. S. Attanasio (Harper and Row: New York, 1970). The *Oxford English Dictionary* gives the first known use of the word “technology” in English as 1615.
- 2 David Noble suggests that Jacob Bigelow, a Boston physician who helped to found the Massachusetts Institute of Technology (1865), was “instrumental” in introducing “technology” into general usage (in its modern sense). To establish this, Noble offers this quotation from Bigelow’s *Elements of Technology* (1829): “There has probably never been an age in which the practical applications of science have employed so large a portion of talent and enterprise of the community, as in the present. To embody...the various topics which

belong to such an undertaking, I have adopted the general name of Technology, a word sufficiently expressive, which is found in some of the older dictionaries, and is beginning to be revived in the literature of practical men at the present day. Under this title is attempted to include an account...of the principles, processes, and nomenclatures of the more conspicuous arts, particularly those which involve the application of science, and which may be considered useful, by promoting the benefit of society, together with the emoluments of those who pursue them." David Noble, *America by Design* (Alfred A. Knopf: New York, 1977), pp. 3-4. My reading of this passage does not agree with Noble's. Bigelow refers to a revival of the term already beginning; hence, by his own admission, he probably is not "instrumental" in its revival. Further, Bigelow's own use of the term is barely distinguishable from that found in the old dictionaries he refers to. The chief difference is the observation that *some* of the more "conspicuous arts" he will describe "involve the application of science". He still seems pretty far from the modern idea of technology as either inventions or the "science of invention".

- 3 See, for example, Plato's *Theaetetus* (III: 172-173): "[A] philosopher is a gentleman, but a lawyer is a servant. The one can have his talk out, and wander at will from one subject to another, as the fancy takes him...[but] the lawyer is always in a hurry..."
- 4 Hannah Arendt, *The Human Condition* (Doubleday: Garden City, New York, 1959), p. 323n.
- 5 "If we assume that the Middle Ages ended with the fifteenth century, then a simple count of inventions made or adopted by Europeans during the period confirms that it was, as regards technics, more creative than any previous epoch in recorded history." *Dictionary of the History of Ideas: Studies of Selected Pivotal Ideas*, edited by Philip P. Weiner (Charles Scribner's Sons, Publisher: New York, 1973), vol. IV, p. 359 (D.S.L. Cardwell, "Technology"). I am, of course, still comparing Greece's Golden Age with a similar stretch of time during the Dark Ages. Were I to compare the Dark Ages with the Hellenistic Period, I would hedge my claim a bit (and begin to worry about how to count inventions).
- 6 See, for example, Spencer Klaw, "The Faustian Bargain", in *The Social Responsibility of the Scientist*, edited by Martin Brown (Free Press: New York, 1971), pp. 3-18.
- 7 The modern prejudice against manual labor seems to vary from place to place and time to time. It is, for example, certainly less in the United States today than, say, in France in the nineteenth century. So, for example, it is today hard to imagine the events a French mechanical engineer (or mechanic) of the nineteenth-century recalled. After church, he struck up a conversation with a young woman (with her mother standing by). When she found out that he built steam engines for a living, she shuttered: "What! You work, you are therefore exposed to all the filth that trade includes?" A bit vexed I responded "But yes, miss, and I dare to believe that none is apparent at this moment." The mother turned her back and the eyes of my beautiful neighbor fell on my well-ground hands, which did not betray me, and she moved away. For her, I was a plague-stricken person." Quoted from Eda Kranakis, "Social Determinants of Engineering Practice: A Comparative View of France and America in the Nineteenth Century", *Social Studies of Science* 19 (February 1989): 5-70, at 13. The whole paper is well worth reading both for the contrast it draws between French and American practice and for the cultural explanation it offers.
- 8 *World Almanac* (World Almanac: New York, 1989), p. 158. This number must, however, be taken only as a rough approximation. The Labor Department (three years later) set the number of engineers at 1,519,000 (*Occupational Outlook Handbook*, U.S. Department of Labor, Bureau of Statistics, May 1992, p. 64); while the National Science Foundation put the number at 2,849,800 (*U.S. Scientists and Engineers: 1988 Estimates, Surveys of Science Resources Series*, National Science Foundation, NSF 88-322, p.6).
- 9 The operative word here is "seems". My evidence is quite scanty: but, given the general invisibility of engineers, the difficulty of counting inventions (as against, say, counting patents), and the absence of any better evidence, it will have to do. Here is my evidence: All the people I know who hold patents are engineers; that, except in industrial design and "genetic engineering" (the scare quotes indicating the almost complete absence of engineers from this field of "engineering"), most inventions reported in the newspaper, even many attributed to "science", seem to have been the work of one or more engineers. That also seems to be true of

the inventions described in the TV series *Inventions*: where the program gives enough biographical detail to determine what an inventor did for a living, he (or she) generally turns out to be an engineer. I readily admit that it would be good to have better evidence than this. Indeed, I urge social scientists to try to get better evidence.

- 10 "Principles of Medical Ethics", in *Codes of Professional Responsibility*, edited by Rena A. Gorlin (Bureau of National Affairs: Washington, DC, 1986), p. 99.
- 11 In the eighteenth century, "profession" was (more or less) a synonym for "calling", a way of making a living. The "liberal professions" were those suitable for a "free man", that is, a gentleman (just as the liberal arts were arts suitable for a gentleman). Generally, the liberal professions required book learning (though the profession of arms seems to have been an exception). The other professions (the "illiberal" ones) would have included crafts like carpentry, tailoring, and tinning, and even "waiting on trade", for example, in a grocery or druggist shop. Of course, the history of "profession" goes back to the middle ages, to the vow one took to enter a holy order. But that is another story, one the professions (in the modern sense) revived as they took form.
- 12 The first of these modern professions was the apothecaries, who reorganized in 1815. The other liberal professions followed only slowly, beginning with the solicitors in the 1830s. See, W. J. Reader, *Professional Men: The Rise of the Professional Classes in Nineteenth-Century England* (Basic Books: New York, 1966), especially, pp. 51-55.
- 13 Thomas Percival, *Medical Ethics; or A Code of Institutes and Precepts Adapted to the Professional Conduct of Physicians and Surgeons* (1803). The word "institutes" suggests that Percival's model here is (in part at least) jurisprudence. (Since the emperor Justinian's famous textbook, "institutes" has signaled that the book so titled was a textbook or summary of the law, the rest of the title telling the particular jurisdiction, for example, Coke's *Institutes of the Laws of England*.) Percival in fact makes this connection in his introduction, indicating that he originally intended to call his text "medical jurisprudence". About half the text is a summary of English law a physician should know. Much of the rest consists of "precepts" (that is, advice) rather than of standards of conduct ("ethics" strictly so called). Medical ethics (in the modern sense) is actually a small part of that seminal work.
- 14 See, for example, M. David Burghardt, *Introduction to the Engineering Profession* (HarperCollins Publishers: New York, 1991), p. 26: "We shall assume that wherever there was an invention or innovation, engineering was required." Burghardt does not say the same about engineers. Can there be engineering without engineers? The same, unfortunately, is true of Billy Vaughn Koen's generally astute *Definition of the Engineering Method* (American Society of Engineering Education: Washington, DC, 1985). Note, for example, p. 26: "After 20 or 30 centuries, the engineer learned how to correct this problem by allowing the front axle to pivot on a king bolt as stage three in the evolution of cart design."
- 15 See, for example, Ralph J. Smith, *Engineering as a Career*, 3rd (McGraw-Hill: New York, 1969), p. 22: "It has been said that the history of civilization is the history of engineering. Certainly it is true that the highly developed civilizations have all been noted for their accomplishments in engineering." Substitute "building" for "engineering" and there would be nothing to object to. The same is true of scholarly works such as Donald Hill, *A History of Engineering in Classical and Medieval Times* (Croom Helm: London, 1984). A careful researcher and writer, Hill actually argues for his application of "engineer" to ancient builders (rather than just assuming it as most writers do). Yet he soon admits that "classical and medieval engineers did not have a quantified, scientific basis for their designs" (p.5), that they lacked formal training characteristic of modern engineers, and that they even lacked a full-time occupation (p. 7). They did "engineering" (read "building") as a sideline. So, whatever they were, they were not a profession—or even an occupation.
- 16 Though we now translate "*corps du génie*" as "corps of engineers", there is in fact no exact English equivalent. While "*génie*" corresponds to the English "gin"(as in "cotton gin") and perhaps "jenny" (as in "spinning jenny"), the French has less suggestion of "engines". Perhaps the best translation would be "corps of the contriver"—though this lacks the suggestion of magic (as in the English "genie"). Unlike the *corps du sappeur* ("the corps of spaders"), for

- example, the *corps du génie* seems not to have taken its name from the implements it used but to have given those implements its name. This suggests an inherent novelty in what it did.
- 17 See, Frederick B. Artz, *The Development of Technical Education in France, 1500-1850* (MIT Press: Cambridge, Massachusetts, 1966), p. 48; or W.H.G. Armytage's (misnamed) history of engineering in Britain, *A Social History of Engineering* (Faber and Faber: London, 1961), pp. 96 and 99.
 - 18 There have, of course, been many changes in the engineering curriculum (as well as many experiments with less demanding curriculum). Whole new subjects, such as thermodynamics or electricity, have been added, and geometry and trigonometry have given way to a second year of calculus. For engineers, these may appear more than improvements in detail—and, for many purposes, they certainly are. Yet, for our purposes, such differences between the French curriculum of 1794 and today's typical engineering curriculum hardly affect the gap between what engineers learn and what lawyers, physicians, or even architects learn. Those who wish to understand what distinguishes engineering from other professions must pay attention to what engineers have in common which they do not share with others, even over long stretches of time, rather than what divides them.
 - 19 Artz, 47-48. I should, perhaps, warn that the members of this corps do not seem to have been known as "civil engineers" (*ingénieurs civils*) but as "road and bridge engineers". The French seem to have reserved "civil engineer" for engineers employed by private persons. All members of a corps, whether of the *corps du génie (militaire)* or of the *corps des ponts et chaussées* were, of course, state employees. The English term "civil engineer" may have derived from a misunderstanding of the French term, since the English, with a relatively weak state, had no exact counterpart to the *corps des ponts et chaussées*. Compare Kranakis, especially, pp. 29-30.
 - 20 Engineers also had some secret methods (for example, Monge's descriptive geometry). Artz, 106.
 - 21 Artz, 81-86.
 - 22 But note Peter Michael Molloy's remark in *Technical Education and the Young Republic: West Point as America's École Polytechnique* (Brown University, Dissertation, 1975), p. 105: "From a description of the curriculum, there should be no mystery for the change in the School's name in 1795 from *École des Travaux Publics* to *École Polytechnique*." It remains a mystery to me.
 - 23 Artz, 154-155.
 - 24 Artz, 160. I should, perhaps, say "this *École Polytechnique*". By 1830, the *École Polytechnique* had become so devoted to mathematics that (by American standards of the day) its graduates generally did not practice engineering. This later *École Polytechnique* may, then, provide an early example of the ability of science education to crowd out engineering. See Molloy, esp., 119-130. On the other hand, this interpretation may simply be unfair (within the French environment). See Kranakis, esp. pp. 22-29, for reasons to think that the *École Polytechnique* remained an engineering school throughout the nineteenth century.
 - 25 Artz, 160-161. I should perhaps add that only practical difficulties seem to have prevented all this from happening as early as 1802. Molloy is very good on this.
 - 26 Eugene Ferguson, "The Imperatives of Engineering", in John G. Burke et al., *Connections: Technology and Change* (Boyd and Fraser: San Francisco, 1979), 30-31. Ferguson's "imperatives" are, of course, Koen's "heuristics".
 - 27 For a detailed study of one of these proxy measures that, in the end, had to be discarded, see Walter G. Vincenti's discussion of "stability", in *What Engineers Know and How They Know It* (Johns Hopkins University Press: Baltimore, 1990), pp. 51-108.
 - 28 In the absence of government action, engineering societies have developed codes, both to enhance efficiency (by promoting standardization) and to maintain safety. These both set standards for engineers to follow "voluntarily" and provide "model codes" legislatures could adopt. For more on one of these model codes, Paula Wells, Hardy Jones, and Michael Davis, *Conflict of Interest in Engineering* (Kendall/Hunt: Dubuque, 1986). Since this standard-setting has been criticized as a usurpation of a governmental function, Ferguson's complaint that engineers should have done more of it seems unfair—at least until he offers a theory of which activities belong to government and which to private organizations or individuals.

- 29 Not all engineering designs dumb-down a job. Many engineering designs “automate”, that is, eliminate the routine work of many while creating technically sophisticated jobs for a few. These may be regarded as the limiting case of dumbing-down or as an entirely different way of doing without large numbers of highly skilled workers. I don’t think much turns on how it is categorized.
- 30 For more on the military connections of engineering, together with the connections between the military and technology, see Barton Hacker’s useful summary, “Engineering a New Order: Military Institutions, Technical Education, and the Rise of the Industrial State”, *Technology and Culture* 34 (April 1993): 1-27.
- 31 Artz, 162.
- 32 Accreditation Board of Engineering and Technology, Code of Engineering Ethics, first principle (1985). Italics mine.
- 33 The first four of Ferguson’s imperatives might also count as ways that engineering differs from science (and most other professions) as well.
- 34 Compare Koen, esp. 63-65. Koen (rightly) points out that engineers sometimes go beyond science (what we now know) and sometimes ignore science (because the client may not be able to afford scientific accuracy) and so at least part of the time cannot be said to be applying science. This is, I believe, an important point, but one quite distinct from the one I am making. Perhaps engineering differs from applied science in enough ways that the interesting question is not how engineering differs from applied science but why they were ever thought to be the same.
- 35 All “exceptions” have been in the electrical engineering department. Whether they are truly exceptions is a matter about which I remain uncertain. On the one hand, these professors of engineering were educated as engineers; on the other, they considered their location in an engineering department to be an accident. They admitted—indeed, declared—they could just as well, or even better, have been lodged in a physics department. They seem, therefore, to have lost their identity as engineers along with their interest in helping to make something useful. In that respect at least, they constitute evidence for, rather than against, my claim.
- 36 Compare Vincenti, 161: “Engineers are after a theory they can use for practical calculations...To obtain such a theory they are willing, when necessary, to forgo generality and precision...and to tolerate a considerable phenomenological component. Scientists are more likely to be out to test a theoretical hypothesis...or infer a theoretical model.”
- 37 For some historical background on this use, as well as my rationale for preferring this definition of ethics, see Michael Davis, “The Ethics Boom: What and Why”, *Centennial Review* 34 (Spring 1990): 163-186.
- 38 For more on this point, see Michael Davis, “Codes of Ethics, Professions, and Conflict of Interest: A Case of an Emerging Profession, Clinical Engineering”, *Professional Ethics* 1 (Spring/Summer 1992): 179-195.
- 39 Fundamental Canon 1, “Code of Engineering Ethics”, Accreditation Board of Engineering and Technology, adopted by most other American engineering societies. For those tempted to think of this provision as somehow “embodying” the whole of engineering ethics, it is worth recalling that this code (and most others) contain many provisions concerned with protecting the client, employer, other members of the profession, or the profession itself. Some codes may soon include provisions protecting the environment and non-humans as well. The paramountcy provision is first among equals, not the mother of all.
- 40 American Institute of Electrical Engineers, *Code of Principles of Professional Conduct* (March 8, 1912).
- 41 Morris Llewellyn Cooke, “Ethics and the Engineering Profession”, *Annals of the American Academy of Political and Social Science* 101 (May 1922), 69.
- 42 Sec. 4, “Canons of Ethics for Engineers”, Engineers’ Council for Professional Development (October 25, 1947).
- 43 1.1 “Canons of Ethics of Engineers”, Engineers’ Council for Professional Development (Ethics Committee, 1963).

Henry Petroski, “Scientists vs. Engineers,” in *An Engineer’s Alphabet: Gleanings from the Softer Side of a Profession* (Cambridge University Press, 2011), 280-283.

scientists vs. engineers. A distinction between scientists and engineers is not always easily or clearly made. Distinctions that are drawn are sometimes clouded by the fact that scientists can do engineering, as so many physicists did during the Manhattan Project, and engineers can do science, as they do when they conduct experiments to collect data that is essential to proceeding with their designs or seek to understand principles behind inventions that obviously work but for reasons that are not fully transparent.

There have been many attempts to clarify distinctions between scientists and engineers, and perhaps the most widely (and variously) quoted is that attributed to the aerospace engineer and scientist Theodore von Kármán (1881–1963): “Scientists seek to understand what is;

engineers seek to create what has not yet been.” Similar sentiments have been expressed in aphorisms galore, among them: “Scientists understand the world, but engineers make it work”; “Scientists investigate what is; engineers create what never has been”; “Scientists seek to know; engineers to do”; “Scientists make it known; engineers make it useful.” Another variation on the same theme, adopted by the Engineers Joint Council, reads: “Engineers plan, design, produce, maintain, and operate. Scientists make it known. Engineers make it useful.” Couched in science/engineering terms, the von Kármán aphorism can be expressed as: “Science makes things known; engineering makes things work.” Gordon S. Brown (1907–1996), who served as dean of engineering at MIT, said of engineers primarily engaged in research that, “they may work as scientists, but their knack of seeing the useful rather than searching for the unknown characterizes them as engineers.”

Another approach to making the distinction between scientists and engineers (and between science and engineering) is to emphasize not what they do but why they do it. According to one observer, “An engineer is a person who will learn an obscure physical law in order to build a complicated piece of technological apparatus. A scientist is a person who will learn how to build a complicated piece of technological apparatus in order to learn an obscure physical law.”

It is a common lament among engineers that all too often in the news media successful technological endeavors and achievements are attributed to science and scientists, whereas technological problems and failures are blamed on engineering and engineers. Thus, landing astronauts on the Moon was hailed as a scientific achievement, but when a test rocket exploded on the launch pad it was described as an engineering failure. This false dichotomy was evident, for example, in the Mars Pathfinder Mission of 1997.

When the Pathfinder landed and celebration erupted in the control room, the participants were identified by commentators as space scientists, whereas in fact they were virtually all engineers. When some communications problems later erupted, television viewers were told that the engineers were working on them. See “Making Headlines,” *American Scientist*, May–June 2000, pp. 206–209; see also *The Essential Engineer: Why Science Alone Will Not Solve Our Global Problems* (New York: Knopf, 2010).

In the wake of the explosion on the *Deepwater Horizon* oil rig and the subsequent months-long oil well leak in the Gulf of Mexico in 2010, there were many instances of action and reporting in which the roles of engineers and scientists were confused, thus giving the lay public misleading caricatures of both. At one point, the Secretary of Energy, a Nobel-prize winning physicist, was sent to the oil-spill command center in Houston with orders from the White House to direct what clearly should have been an engineering effort under the direction of an engineer to cap the leaking well. In the end, it was not scientific knowledge and achievement that ended the gushing but engineering experience and technological savvy. In one newspaper report describing the political decision in the wake of the accident to impose a moratorium on all deep-water drilling, consulting engineers were repeatedly referred to as scientists throughout the entire first two-thirds of the rather long story on their report as submitted, which did not support a ban. It was only after the consultants were identified as having been recommended by the National Academy of Engineering that they were correctly described as engineers – but only for a few paragraphs before the reporter lapsed back into calling them scientists. See Kara Rowland, “Anger Overflows on Drilling Halt Report,” *Washington Times*, November 10, 2010.

Blaming the engineer is not a modern phenomenon. In his first-century A.D. treatise on the water supply system

of Rome, the water commissioner Frontinus described an incident that he experienced at the site of the excavation for a tunnel to carry water to the Algerian seaport of Saldae: “There I found everybody sad and despondent; they had given up all hopes that the two opposite sections of the tunnel would meet, because each section had already been excavated beyond the middle of the mountain, and the junction had not yet been effected. As always happens in these cases, the fault was attributed to the engineer, as though he had not taken all precautions to insure the success of the work.”

According to Frontinus, in the absence of the engineer “the contractor and his assistant had made blunder after blunder, in each section of the tunnel they had diverged from the straight line, each towards his right and had I waited a little longer before coming, Saldae would have possessed two tunnels instead of one.” The situation was corrected by doing a further survey and digging a transverse tunnel that connected the two diverging ones. See Frontinus, *The Water Supply of the City of Rome*, translated by Clemens Herschel (Boston: New England Water Works Association, 1973), pp. 151–152.